

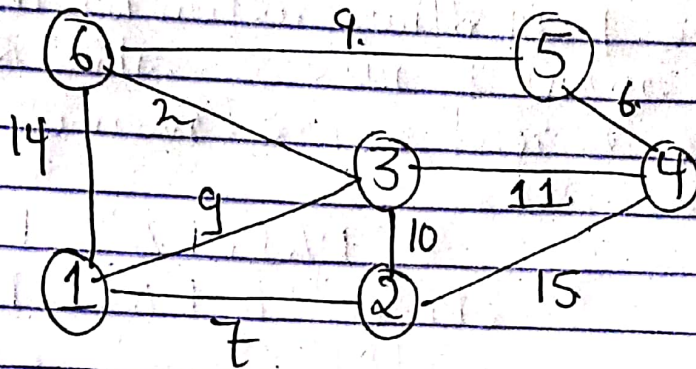
Dijkstra Algorithm → Single source shortest path problem

→ Only one source is given
 17 Real life example → Google Map

* Relaxation if $d(u) + c(u,v) < d(v)$
 $d(v) = d(u) + c(u,v)$

* Dijkstra works well for both directed and undirected

Eg 1



* find min from table

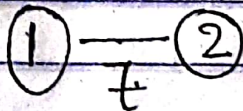
Source 1

Src 1

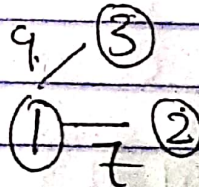
Destination

	2	3	4	5	6
1	7	9	∞	∞	14
2	7	9	22	∞	14
3	7	9	20	∞	11
4	7	9	20	20	11
5	7	9	20	20	11

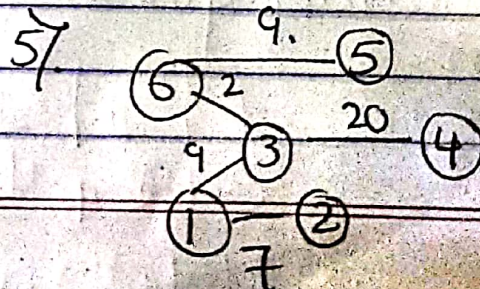
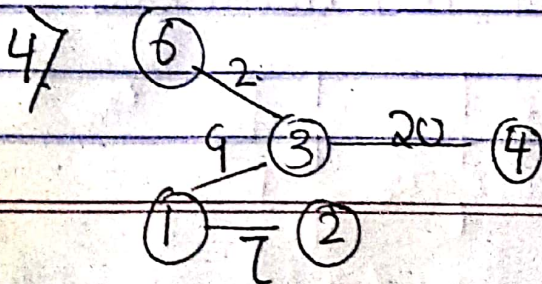
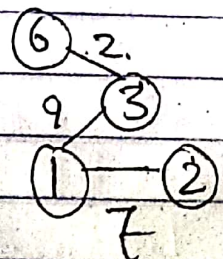
17



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Algorithm

// Algo Dijkstra (G, S)

// Input: Graph with weights, source

// Output: Shortest dist from S to all other vertices

Start $(V) \rightarrow (V, \infty)$

1. for each vertex V
 $dist[V] = \infty$

$visited[V] = False$

2. $dist[S] = 0$

3. While there exist unvisited vertex

$u =$ Vertex with min dist among unvisited vertex

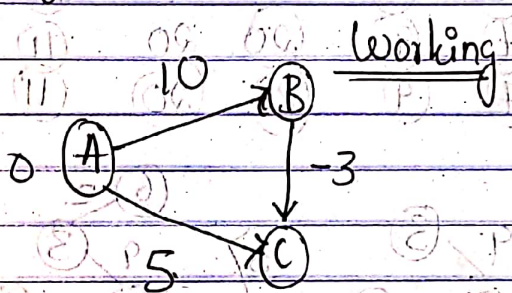
$visited[u] = True$

for each neighbor v of u

if $dist[u] + weight(u, v) < dist[v]$:
 $dist[v] = dist[u] + weight(u, v)$

End

* Effect on -ve weight Edge.



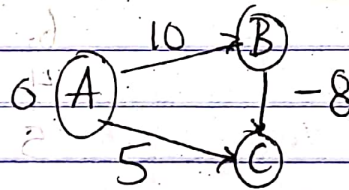
A { B C

$\infty \infty$

C { 10 5

B { 10 5

Not working



A { B C

$\infty \infty$

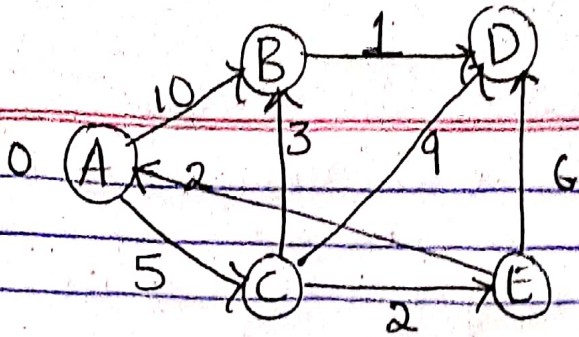
A, C { 10 5

A, C, B { 10

but 5 is already

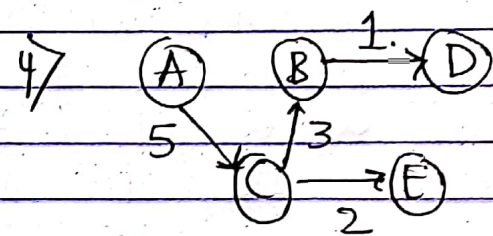
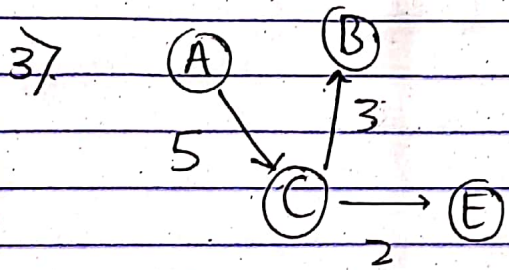
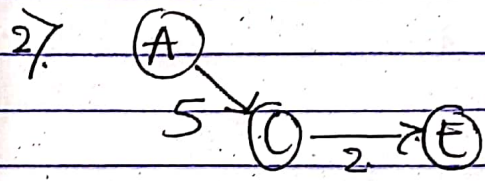
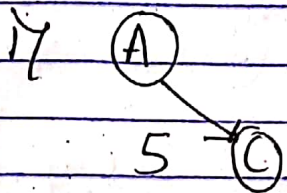
relaxed

Eg 2

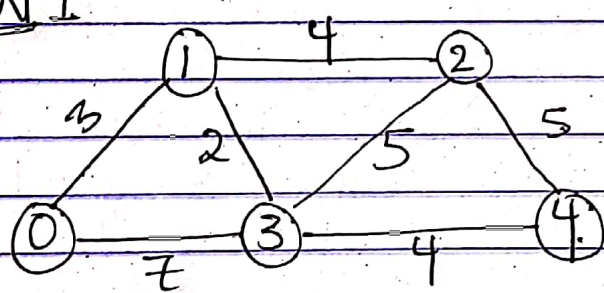


Sr/A

	B	C	D	E
A	∞	∞	∞	∞
C	10	(5)	∞	∞
E	8	(5)	14	(7)
B	(8)	(5)	13	(7)
D	(8)	(5)	(9)	(7)



H.W 1



H.W 2

